Human-in-the-Loop Legal Named Entity Annotation with LLM Assistance

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Abstract. High-quality Named Entity Recognition (NER) in the legal domain is often hindered by the scarcity of reliable datasets and the structural noise inherent in legal corpora. To address this, we present a semi-automated Human-in-the-Loop (HITL) architecture designed for the rapid development/enhancement of legal NER datasets. Our pipeline integrates a hybrid named-entity extraction engine, combining heuristic-based approaches with a fine-tuned Transformer, and a semantic similarity module within an active learning loop to facilitate named-entity annotations. Crucially, a Large Language Model (LLM) framework acts as a neural adjudicator assistant to resolve conflicts and enforce strict entity constraints. We demonstrate the usefulness of the proposed method to build an improved version of the Greek Legal NER dataset expanding the named entity volume by 209% while reducing manual annotation to merely 1.4%. Finally, we report evaluation results using the new dataset and a diverse set of NER methods under fully supervised and zero/few-shot scenarios.

Keywords: Legal Information Extraction · Named Entity Recognition · Human-in-the-loop · Active Learning

1 Introduction

Named Entity Recognition (NER) is essential for legal Natural Language Processing (NLP) downstream tasks such as contract analysis and compliance checking [3, 15]. Despite advances in Large Language Models (LLMs) [2, 30], the primary bottleneck in legal NLP remains the scarcity of high-quality training data [29].

For Greek legal NLP, the Greek Legal NER (GLN) dataset [1] has served as the de facto standard and it is included in the multilingual benchmark LEXTREME used to evaluate legal NLP approaches [16]. However, a careful examination reveals that this legal resource suffers from critical structural flaws, including severe annotation sparsity, rigid single-category mappings, and fragmented entity boundaries stemming from early PDF-to-text extraction pipelines (see Table **??**). Models trained and evaluated on such noisy datasets inevitably propagate these artifacts, rendering extracted conclusions unreliable [17, 24].

Manually correcting such large-scale systemic errors across thousands of legal documents is prohibitively expensive. Furthermore, while LLMs, Active Learning

Table 1: Visual comparison of annotation quality. GLN v1 severe sparsity, truncated spans, and context errors. The Enhanced GLN v2 through the restoration architecture, resolves these issues.

Dataset Version	Annotated Sequence Visualization (Translated to English)
Example A: Annotation Sparsity & Boundary Truncation	
Legacy GLN v1	According to Decision 707/2011$_{\text{LEG-REFS}}$ "Authorization of Operations", the Council of State convened in Athens.
Enhanced v2	According to Decision 707/2011 "Authorization of Operations"$_{\text{LEG-REFS}}$, the Council of State$_{\text{ORG}}$ convened in Athens$_{\text{LOC}}$. ↪ *Fix: Overcame single-entity bias, recovered missing ORG & LOC, expanded document title.*
Example B: Context Conflation, DATE addition & Spilt Entities	
Legacy GLN v1	Athens$_{\text{GPE}}$, 12 May 2016, the Ministry$_{\text{END OF SENTENCE}}$.
Enhanced v2	Athens$_{\text{LOC}}$, 12 May 2016$_{\text{DATE}}$, the Ministry of Labour$_{\text{ORG}}$. ↪ *Fix: Resolved LOC vs GPE conflation, introduced DATE entity, restored sentence splitting errors.*
Example C: Multiple Entity Capture Failure	
Legacy GLN v1	Pursuant to the provisions of Law 3044/2002$_{\text{LEG-REFS}}$, Law 3852/2010 and Presidential Decree 133/2011.
Enhanced v2	Pursuant to the provisions of Law 3044/2002$_{\text{LEG-REFS}}$, Law 3852/2010$_{\text{LEG-REFS}}$ and Presidential Decree 133/2011$_{\text{LEG-REFS}}$. ↪ *Fix: Resolved failure of capturing all contiguous same entity types.*

(AL) and Multi-Agent approaches serve as capable annotation assistants [9, 22, 7], they often struggle to enforce the rigid constraints required in legal texts and avoid hallucinations [18].

To overcome this, we design a Human-in-the-Loop (HITL) methodology ensuring high quality of legal named entity annotation. Annotation suggestions are provided by heterogenous methods, including simple rule-based methods and fine-tuned Transformer models. To resolve conflicts (e.g., overlapping entity boundaries or ambiguous entity types), we use a LLM framework that acts as a neural adjudicator assistant. In addition, a semantic similarity mechanism extracts similar cases in the corpus and propagates verified annotation corrections to facilitate the rapid devolopement of the curated dataset.

We apply the proposed approach to GLN and build an improved version resolving the vast majority of the existing annotation flaws and enriching the annotated named entity volume substantially. We also augment the named entity types by incorporating dates. The new, improved version of GLN is used in legal NER experiments where a diverse set of NER methods are evaluated.

In summary, our main contributions are:

- **A HITL approach assisted by LLMs:** The proposed semi-automated approach applies active learning by repeatedly estimating candidate annotations, resolving conflicts using LLMs, and ensuring high-quality annotations through a HITL process. In addition, we introduce a semantic similarity mechanism that facilitates the rapid annotation of similar cases.
- **An improved GLN dataset:** We systematically resolve annotation flaws in GLN, a dataset widely used in legal NLP evaluations, using the proposed annotation method and release the enhanced version of GLN[1]. This new dataset inlcudes significantly more annotated named entities and we also expand the named entity types incorporating dates.
- **:Baseline benchmarking** We provide an initial evaluation of diverse NER paradigms on the improved GLN dataset, establishing baseline performance.

2 Related Work

2.1 Legal NER Datasets and Paradigms

The extraction of structured information from legal texts is supported by specialized domain datasets, such as CUAD [10] for English contracts, MAPA [8] for EU administrations, and the Indian Legal NER corpus [13].

Since legal documents frequently contain nested entities and extensive multi-token spans—a structural challenge mirrored in biomedical texts [6]—span-based architectures like GLiNER [28] are increasingly necessary. Furthermore, the task of extracting long entity boundaries from extended legal texts has recently driven the development of lightweight, memory-efficient span extractors like SeNER [5]. While advanced representation techniques, such as the Automated Concatenation of Embeddings (ACE) [23], can significantly optimize structured predictions, their real-world efficacy remains fundamentally bottlenecked by the quality of the underlying annotations. Consequently, precise span localization in zero- and few-shot settings has driven the adoption of boundary-modeling techniques (e.g., DiffusionNER [20] and BANER [14]), which offer robust pathways for complex entities, provided the ground-truth data is structurally sound.

2.2 LLM-based Annotation and Multi-Agent Systems

LLMs exhibit zero-shot capabilities [25] comparable to human annotators [9] and are increasingly integrated into AL loops [27]. Decoupling semantic identification from deterministic boundary prediction reduces operational load and hallucinations [21]. Because relying solely on LLMs incurs high costs and boundary inaccuracies, hybrid approaches like UniversalNER [12] combine LLM contextual inference with specialized encoders. To further address hallucinations, multi-agent systems [26, 11] employ collaborative roles, utilizing debate [7] and LLM-as-a-Judge paradigms (e.g., ChatEval [4]) to resolve ambiguities via consensus. However, deploying generalized multi-agent approaches for strict Legal

[1] Availability: `https://anonymous.4open.science/r/NER_GLN_v2_curation_architecture-4742/`

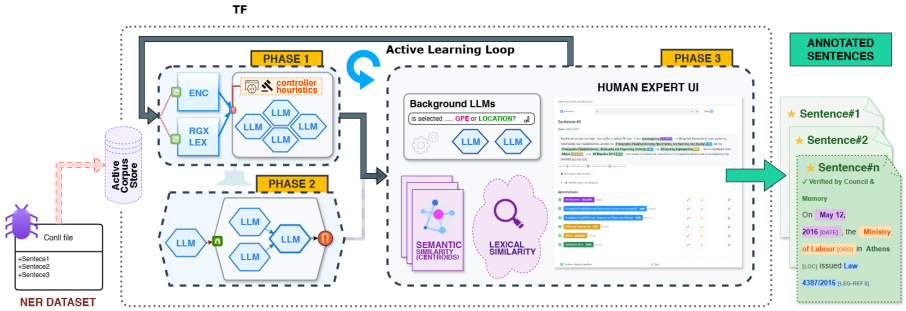

Fig. 1: The proposed annotation methodology. Phase 1 generates annotation suggestions based on various NER methods. Phase 2 combines suggestions via LLM neural adjudication. Phase 3 facilitates HITL annotation verification and exploits, semantic and lexical similarity to propagate annotation to other similar cases.

NER remains challenging: they optimize for high-level generative approximations rather than rigid entity constraints, risking boundary drift. Operating largely as autonomous pipelines, they also lack seamless HITL integration. Consequently, targeted architectures that organically fuse deterministic heuristics with multi-agent neural adjudication are critical to effectively transition the human expert from a manual annotator to a high-level adjudicator, ensuring the structural rehabilitation of broken datasets.

3 The Proposed Annotation Methodology

The proposed method consists of three phases that are repeated in an AL framework: Phase 1 generates annotation suggestions using a set of heterogeneous NER approaches. Phase 2 functions as a neural adjudicator assistant, autonomously resolving conflicts and boundary overlaps. Finally, Phase 3 provides an interactive HITL workspace where human-validated annotations are provided and propagated across the dataset using semantic and lexical similarity mechanisms to retrieve similar cases. LLMs are used in all three phases.

3.1 Phase 1: Annotation Generation

The initial step is to generate annotation suggestions. In order to ensure high recall, we use a hybrid annotation extraction engine combining heuristic-based approaches (i.e., regular expressions and specialized lexicons) with a fine-tuned Transformer backend (`GreekLegalRoberta` [19]). The latter is fine-tuned on a human-annotated part of the dataset. In the case of GLN, we use the test split for that purpose. This part of the dataset is extended in the next cycle of AL with new annotated entities. Additionally, early LLM interventions are deployed in this phase to perform adjustments, such as filtering low-confidence predictions and isolating immediate entity overlaps (e.g., distinguishing an administrative `GPE` from a spatial `LOCATION`).

3.2 Phase 2: Neural Adjudication Assistant

A two-stage process is used to autonomously resolve complex entity boundary conflicts and overlapping entity boundary predictions obtained from phase 1. **Stage 1: Entity-Specific Pre-Auditing.** An LLM performs class-aware triage of Phase 1 annoatation suggestions. Each candidate annotation is classified as Accept, Reject, or Fix; boundaries are recalculated immediately if needed. Confidently accepted items are removed from the conflict pool. **Stage 2: Multi-Agent Debate.** Remaining ambiguities escalate to a multi-agent debate. First, two revision LLM agents re-examine the context of each annotation. Then, a neural judge LLM adjudicates their arguments to resolve ambiguities. **Deterministic Span Alignment:** Gestalt pattern matching is applied to find longest annotation and reduce structural noise before human review.

3.3 Phase 3: HITL and Bulk Annotation

All candidate annotations flow into a HITL collaborative workspace for human review and verification. To combat annotator fatigue, the workspace features dynamic AI tools tailored to correct specific dataset flaws. **On-Demand Boundary Refinement:** For proposed entities with misaligned boundaries, annotators can trigger an entity-specific LLM. Utilizing schema guidelines, the model autonomously recalculates the exact character offsets, instantly snapping the correct boundaries into place. **Zero-Shot Entity Discovery:** To rescue false negatives that eluded both heuristics, the base Transformer and the Phase 2, an LLM can be invoked to perform zero-shot extraction over the raw sentence. This acts as a high-recall fallback, successfully surfacing previously unrecorded entities. **Context-Aware Assistant:** An integrated prompting terminal allows annotators to query an LLM to resolve lingering semantic ambiguities prior to finalization.

3.4 Global Propagation via Vector Memory and Fuzzy Mechanics

Once an annotator manually validates, corrects, or rejects an entity in Phase 3, resolving the identical error across thousands of documents is highly inefficient. To automate this, the architecture incorporates semantic and lexical mechanisms for bulk annotation actions (Accept, Reject, Change Type, or Auto-Fix Boundaries).

Semantic Propagation via Dynamic Centroids: During Phases 1 and 2, the Transformer computes and stores contextual embeddings for all proposals. Upon selecting a validated proposal, its dense vector is retrieved, and cosine similarity is used to find contextually similar entities across the corpus. A *Dynamic Prototype Updating* mechanism recalculates the cluster's centroid as the mean vector of all accepted instances during bulk-acceptance, iteratively refining the search space to capture precise semantic clusters (e.g., diverse variations of `ORG`).

Lexical Propagation and Usage Scenarios: The system employs a fast lexical fallback—substring and fuzzy matching—implemented via RapidFuzz.

This parallel layer complements vector retrieval: when an entity follows a rigid structural template (e.g., boilerplate legal references or OCR-noisy strings like "yp' arithm."), RapidFuzz computes Levenshtein-based similarity scores. Matches above a cetrain threshold are treated as practical exacts. By operating in concert, an annotator can use fuzzy matching to retrieve structural variations of `PUBLIC-DOCS`, execute bulk acceptance, and optionally trigger the LLM Boundary Refinement tool to globally correct exact character offsets in a single workflow.

4 Dataset Analysis and Targeted Rehabilitation

In this section, we present the empirical analysis of the Greek Legal NER (GLN) v1 dataset and its targeted rehabilitation into the Enhanced GLN v2. We demonstrate how the legacy corpus suffered from critical structural flaws and how our customized algorithmic tools and HITL interfaces were specifically deployed to resolve these exact anomalies, quantitatively transforming the resulting dataset.

4.1 Exposure of Legacy Flaws in GLN v1

A rigorous preliminary audit of the GLN v1 baseline revealed pervasive systemic errors that severely compromise its reliability as an evaluation benchmark. The legacy corpus suffered from profound **annotation sparsity**, frequently yielding only one annotated entity per sentence. Furthermore, the dataset exhibited **rigid single-category mappings**, systematically conflating distinct semantic concepts (e.g., tagging all instances of "Athens" strictly as a Geopolitical Entity `GPE`, even when used purely as a spatial `LOCATION`). Most critically, the corpus exhibited **structural artifacts** primarily driven by erroneous sentence splitting. Segmentation mechanisms arbitrarily bisected continuous text, fracturing multi-word entities across separate sentences.

4.2 Targeted Algorithmic Interventions

To combat the specific pathologies of GLN v1, our multi-tier architecture was deployed as a targeted rehabilitation engine. System database logs tracking the algorithmic interventions demonstrate the system's capacity to systematically resolve the corpus's critical ontological bottlenecks:

Ambiguity Resolution: To fix the inherent GPE vs. LOCATION conflation in Greek administrative texts, the multi-agent system successfully intercepted and correctly resolved 2,557 cross-label disputes. It leveraged contextual neural adjudication to decouple authoritative actors from purely spatial references.

Conflict Adjudication: For spans where programmatic extraction methods disagreed with the neural base models (often due to PDF fragmentation), the system functioned as the ultimate adjudicator, aligning boundary and label

overlaps in 1,597 structural conflicts concerning `LEG-REFS` (Regex vs. RoBERTa collisions).

Zero-Shot Schema Expansion: Although the underlying legal texts contained numerous temporal references, the legacy schema suffered from temporal blindness. To fix this, our hybrid extraction engine facilitated the introduction of the novel `DATE` category from a cold start, autonomously identifying 2,183 instances without requiring any human seed examples.

Deep Auditing and Target Refinement: Operating as a rigorous quality filter against legacy noise, the Phase 2 Neural Adjudicator proactively removed 13,515 false positives (contextual noise), structurally refined the exact token boundaries of 7,178 entities, and discovered 9,905 previously unrecorded entities that raw heuristics had missed.

4.3 Minimization of Human Cognitive Load in Corpus Rescue

While algorithmic models provided the heavy lifting, the creation of the ground truth required human validation. Out of the finalized **28,986 valid annotations** of the Enhanced GLN v2, human experts were required to perform isolated, de novo manual annotation (drawing entities entirely "from scratch" to identify missing data) in only **1.4%** of the cases (481 manual additions). This drastic reduction in manual cognitive load was driven by tool-assisted mechanics tailored for Greek legal texts:

- **Memory-Augmented Propagation:** The Vector Memory Agent tracked human adjudications in real-time. Whether validating recurring administrative entities, modifying boundaries, or rejecting systematic false positives, the system globally propagated these actions. This autonomous propagation successfully resolved ∼**46%** of the initial curation workload with zero repetitive manual input.
- **Fuzzy Retrieval Logic:** To explicitly combat legacy OCR errors and administrative boilerplate prefixes (e.g., varying abbreviations of "yp' arithm."), annotators utilized the Fuzzy Search agent. This subsystem enabled the rapid mass execution of targeted accepts, rejects, and corrections, successfully resolving ∼**31.8%** of the corpus that strict Regex or dense embeddings failed to capture.

4.4 Quantitative Enhancements and Evaluation Splits

The cumulative effect of the proposed targeted rehabilitation is evident in the dramatically enhanced density and structural integrity of the resulting corpus (Enhanced GLN v2).

Recall Improvement and Boundary Restoration: The total entity count in the Training set increased exponentially from 9,368 to 28,986 (an approximate 209% increase), systematically eradicating legacy false negatives, such as overlooked administrative actors, while concurrently enriching the dataset through the introduction of a novel `DATE` entity to capture previously ignored temporal references. To address previous annotation spans, the pipeline's overlap merging mechanics were deployed. Consequently, the average entity length increased

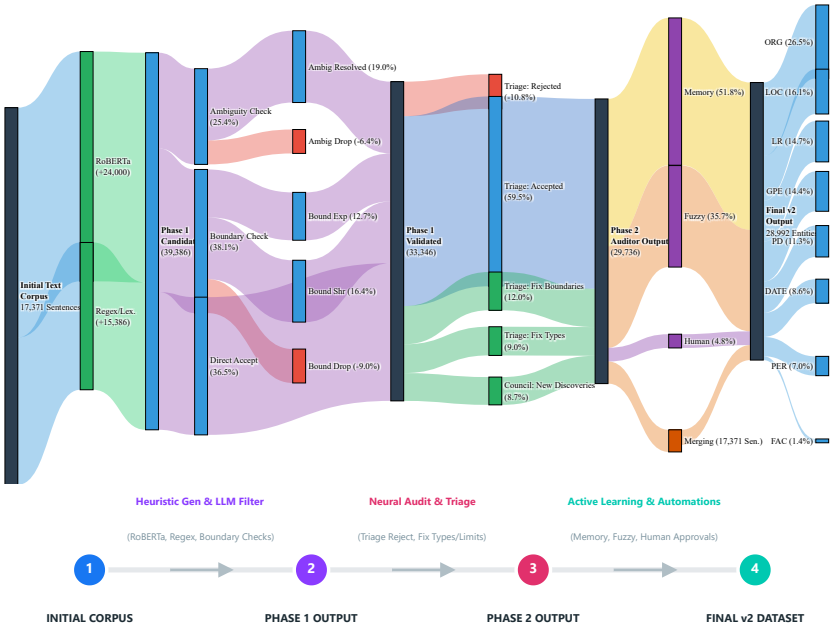

Fig. 2: The chronological transition visualizes the quantitative entity lifecycle from hybrid candidate generation (Phase 1), through multi-agent neural adjudication (Phase 2), to the final HITL verification and global propagation mechanics (Phase 3).

from 3.58 to 4.88 tokens (a 36% increase). This expansion is particularly vital for complex legal citations; rather than isolating bare numerical identifiers (e.g., merely extracting "Law 3907/2011"), the corrected boundaries now encapsulate the full descriptive topics and titles (e.g., "Law 3907/2011 on the Establishment of the Asylum Service"). Furthermore, utilizing the platform's sentence-merging utility, annotators corrected systemic sentence-splitting artifacts, condensing the Training set from 17,699 to 17,371 continuous sentences. This structural reunification successfully restored multi-word entities—such as disjointed acronyms or Ministry names arbitrarily fractured across separate lines—ensuring they are processed as cohesive semantic units.

Semantic Disambiguation and Schema Alignment: Beyond expanding boundaries, the targeted rehabilitation resolved deep-rooted contextual conflations. For instance, the legacy corpus systematically mapped occurrences of city names (e.g., "Athens") strictly to the `GPE` category, regardless of context. The HITL pipeline explicitly decoupled these representations, enforcing a strict functional distinction between physical spatial environments (`LOCATION`) and actual administrative or governing authorities (`GPE`), thereby eliminating conflicting training signals.

Curation Strategy for Uncompromised Benchmarking: To evaluate the rehabilitated dataset without algorithmic bias, we separated the validation

methodology. The **Training Set** and **Development Set** were processed utilizing the full spectrum of the automated platform to rapidly generate a dense, high-quality learning signal. Conversely, to ensure an uncompromised ground-truth benchmark, the **Test Set** was subjected to independent manual adjudication. The final validation on the test set bypassed bulk-automation tools and relied on strict manual curation, producing a highly reliable Gold Standard for subsequent baseline evaluations.

5 Experimental Evaluation and Benchmarks

To rigorously evaluate the structural integrity and complexity of the Enhanced GLN v2 dataset, we benchmarked it across diverse Named Entity Recognition paradigms (Table 2). We compared traditional token-classification Transformers with modern span-based and generative architectures under fully supervised, zero-shot, and few-shot scenarios. This comprehensive setup allows us to assess how the dataset's dense schema and extended entity boundaries—such as lengthy legislative references and administrative documents—challenge current state-of-the-art models.

5.1 Token-Based Sequence Labeling

We fine-tuned six prominent encoder-only architectures: DeBERTa, XLM-R Base, GreekBERT, RoBERTa, MiniLM, and DistilBERT. These models formulate NER as a sequence labeling task. DeBERTa achieved the highest global performance, closely followed by the multilingual XLM-R Base and the domain-specific GreekBERT. While these models excel at identifying standard entities like `PERSON` and `DATE`, their performance drops significantly on structurally complex and expansive categories such as `PUBLIC-DOCS` and `FACILITY`. This discrepancy highlights the inherent limitations of standard token-level classification when processing extended legal spans, as it struggles to maintain cohesive boundaries over long sequences.

5.2 Span-Based and Generative Architectures

To address the limitations of token-by-token tagging, we evaluated span-based frameworks, including GLiNER [28], DiffusionNER [20], and the highly memory-efficient SeNER. These architectures are theoretically better suited for capturing the extensive and occasionally overlapping boundaries typical of legal texts. Notably, DiffusionNER is highly competitive on `LEG-REFS`, demonstrating its generative strength in refining continuous boundary coordinates rather than predicting individual tags. However, their global performance trails behind the top token-based encoders, indicating that while span formulations capture specific long entities well, they struggle with the overall semantic density of the corpus.

Zero-Shot Generalization: We tested GLiNER's zero-shot capabilities by prompting it directly with our schema labels. The model's massive general-domain pre-training endows it with robust innate representations for universal

Table 2: Detailed F1-score performance across all models and entity types on the Enhanced GLN v2. The absolute best scores per column are marked in **bold**. Within the Span-based category, the highest scores are underlined. *Abbreviations: LR = LEG-REFS, PD = PUBLIC-DOCS.*

	Global F1		**Entity-Specific F1**							
Model	**Micro**	**Macro**	**PER**	**ORG**	**GPE**	**LOC**	**DATE**	**LR**	**FAC**	**PD**
Sequence Labeling Architectures										
DeBERTa	**71.2**	**67.3**	93.4	57.5	**72.9**	71.0	80.4	79.4	17.8	39.8
XLM-R Base	69.6	66.7	90.0	60.1	69.4	**71.2**	**81.6**	78.8	**28.4**	40.9
GreekBERT	69.4	65.9	**94.3**	59.0	67.4	70.9	75.5	78.0	21.4	47.3
RoBERTa	69.1	65.8	90.1	57.6	67.2	67.4	81.0	78.9	20.0	**48.7**
MiniLM	68.4	63.5	87.7	56.0	65.7	68.1	79.8	**80.1**	8.8	44.0
DistilBERT	65.5	62.1	79.1	52.2	63.7	68.0	78.3	72.8	17.2	44.5
Span-based Architectures										
Fully Supervised										
GLiNER	64.3	61.3	87.3	**61.5**	70.5	60.3	78.0	66.9	20.6	45.4
DiffusionNER	55.9	53.5	76.0	47.3	59.0	46.6	58.0	77.1	18.3	45.3
SeNER	54.9	49.8	83.8	53.7	62.1	69.4	58.8	31.1	0.0	40.0
Few-Shot Learning										
GLiNER (500 samples)	60.5	57.6	81.3	55.8	63.1	63.6	75.1	63.0	20.5	38.5
Zero-Shot Learning										
GLiNER (Zero-shot)	32.0	28.0	44.0	40.0	9.0	42.0	57.0	21.0	9.0	2.0

concepts, allowing it to perform reasonably well on `DATE` and `PERSON`. However, it failed entirely on domain-specific constructs like `PUBLIC-DOCS` and administrative `GPE`s, confirming that general pre-training is insufficient for specialized legal semantics.

Few-Shot Adaptation: In the legal domain, assembling large-scale annotated data for novel, emerging legislative categories is often unfeasible. To evaluate rapid adaptation efficiency, we fine-tuned GLiNER on a heavily restricted subset of just 500 annotated samples. The performance nearly doubled (from 32.0% to 60.5% Micro-F1), while structurally complex classes like `LEG-REFS` surged from 21.0% to 63.0%. This demonstrates that the rehabilitated dataset provides a highly potent learning signal; modern bi-encoders can adapt to complex legal schemas with minimal human supervision, offering a viable alternative when large-scale annotation is restricted.

5.3 Comparative Analysis

The benchmark results reveal a clear dichotomy: robust token-classification models (DeBERTa, XLM-R) remain the most reliable generalists for dense legal NER,

Table 3: Dataset Statistics: Sentence and Entity Counts comparing the legacy v1 and the Enhanced v2 schema. The Automation columns indicate the estimated percentage of entities dynamically retrieved and boundary-adjusted via the Vector Memory and Fuzzy Curation mechanisms. *Note: Location in v1 includes LOC-UNK & LOC-NAT.*

	Train Set		**Dev Set**		**Test Set**		**Automation**	
Metric / Type	**v1**	**v2**	**v1**	**v2**	**v1**	**v2**	**Memory**	**Fuzzy**
Total Sentences	17,699	**17,371**	4,909	**4,751**	4,017	3,879	–	–
ORG	2,545	**7,712**	643	**1,300**	518	**1,774**	55.0%	32.3%
LEG-REFS	1,306	**4,342**	433	**1052**	419	**1,311**	62.8%	35.0%
PUBLIC-DOCS	620	**3,389**	132	**829**	122	**796**	15.5%	84.0%
GPE	3,002	**4,198**	727	**1,245**	586	**828**	20.0%	28.7%
LOCATION	212	**4,685**	34	**217**	40	**707**	35.0%	25.0%
PERSON	771	**2,017**	199	**314**	242	**516**	85.0%	2.7%
DATE	–	**2,509**	–	**502**	–	**553**	65.0%	1.8%
FACILITY	930	**402**	24	**27**	87	**84**	35.0%	6.7%
Total Entities	**9,386**	**29.254**	**2,192**	**5,331**	**2,014**	**6,569**	**46.1%**	**31.8%**

offering the highest overall accuracy. Conversely, span-based and generative models show localized superiority in defining strict boundaries for extensive entities. Furthermore, the steep performance drop across all architectures for administrative categories confirms that our rehabilitated dataset effectively surfaces the persisting challenges of legal information extraction, providing a rigorous, artifact-free testbed for future NLP research.

6 Conclusion & Future Work

In this paper, we presented a comprehensive, semi-automated Human-in-the-Loop (HITL) architecture designed specifically for the rehabilitation of flawed legal NER datasets. By applying this targeted framework to the legacy Greek Legal NER corpus, we successfully transformed a sparse, artifact-ridden baseline into the dense and structurally robust Enhanced GLN v2.

While Multi-Agent Systems exhibit immense potential for autonomous conflict resolution and semantic inference, our deployment highlighted several practical limitations. As of 2026, the computational and API-related costs of deploying neural adjudicators over entire corpora remain prohibitively high for unconstrained processing (often scaling exponentially to thousands of dollars for full-corpus execution, as multi-turn agent debates require vast, repetitive context windows that rapidly consume millions of tokens). Furthermore, fully mitigating generative hallucinations and ensuring absolute character-level boundary precision without deterministic guardrails are still open research challenges. However,

the trajectory of MAS research indicates that these systems will become increasingly efficient and reliable over time.

To offset these neural costs, our architecture demonstrated the efficacy of coupling generative agents with deterministic tools. While external vector memory is extensively utilized during model inference (e.g., RAG architectures), its integration into active learning and semi-automated annotation schemes is still in its infancy. Our findings establish that even a streamlined implementation of dynamic semantic centroids can drastically reduce manual annotation efforts and bypass redundant LLM API calls. Similarly, approximate string matching proved to be an invaluable mechanism in the legal domain; it effectively navigates the repetitive boilerplate terminology, OCR-induced noise, and minor morphological variations characteristic of legal texts, ensuring high recall where dense embeddings might fail.

Future Work Future methodological research will focus on deepening the integration of the Vector Memory module within the active learning loop. We plan to explore a dynamic fusion of semantic and approximate lexical metrics to create a unified propagation score. Additionally, we aim to refine the creation of memory centroids by leveraging attention-weighted embeddings, allowing the memory to focus on the most discriminative tokens of an entity rather than standard mean pooling.

Regarding the dataset, our immediate goal is the continuous refinement of the Enhanced GLN v2 by auditing any residual, edge-case structural artifacts. Another expansion that could be vital, is the injection of the corpus with contemporary Greek legal documents. Because the foundational texts of GLN v1 originate from a specific historical window, incorporating modern legislative and administrative texts will expose the models to evolving legal drafting conventions, newly introduced terminology, and modern stylistic structures, further increasing the dataset's real-world applicability.

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